



LogoKit

This new cybercrime tool changes logos and text on a phishing page in real time to adapt to targeted victims. <http://dgit.in/feb21-43>



Chromebook Fingerprint

Google finally lets ChromeOS users sign-in using a PIN or fingerprint. <http://dgit.in/feb21-44>

DEVELOPMENT OF MILITARY RADARS

In the 1930s, eight countries that were concerned with the prevailing military conditions started experimenting heavily with radar technology. These countries are the United States, Great Britain, Germany, the Soviet Union, France, Italy, Japan, and the Netherlands. Most of these countries already had some form of existing radar equipment in the military service by the start of WWII. Radar technology was actually a crucial element in ending the war sooner and in the Allies' favour.



SCR-270 at Opana, Oahu, that detected the Japanese attack aircraft

In the US, radar technology was faced with a lack of interest by the US Navy officials up until it was learned how to use a single antenna for transmitting as well as receiving signals (now termed as monostatic radar) to detect and track ships and aircraft. A radar system that employed this was first demonstrated on a battleship named USS New York at 1939. In the same year, the US Navy came up with the acronym 'radar', short for '**R**adio **D**etection **A**nd **R**anging'. The first radars developed by the US Army included the SCR-268 (for controlling antiaircraft gunfire) and the SCR-270 (for detecting aircraft). In fact, it was the SCR-270 that detected Japanese war airplanes approaching Pearl Harbor on December 7, 1941.

Great Britain kicked off radar research for military purpose in 1935. In 1938, the first British radar system, Chain Home (operated at 30 MHz), became operational and it remained so

throughout the course of the war. This radar system allowed Britain to deploy its numbered air defences against imposing German air attacks during the first few years of war.

The Soviet Union also began working on and using radar technology during the 1930s. The Soviet developed different types of radar systems. They were also building an aircraft-detection radar that operated at 75 MHz. Germany also progressed rapidly in radar technology development at the beginning of World War II. The Germans employed radar systems both on the ground and in the air for defence against Allied bombers. However, they made the mistake of abandoning radar development in the late 1940s since they believed that the war was reaching its end. The United States and Britain only accelerated their efforts in the field and overtook Germany before the country realised their mistake.

Now, apart from some German radars that operated at 275 and 560 MHz, all other operating radar systems developed before WWII began were in the VHF band. Meaning, they operated at a frequency below 200 MHz. VHF posed several problems. Firstly, VHF beamwidths are broad, therefore, they have poorer accuracy and resolution in comparison to narrow beamwidths. Secondly, the VHF part of the electromagnetic spectrum does not allow for wide bandwidths that are required for short pulses which allow for greater accuracy. Thirdly, VHF is also subject to atmospheric noise, which limits the receiver's sensitivity. Despite these drawbacks, VHF band was heavily used in the 1930s, successfully too.

Microwave radars started coming into prominence at this time. These systems used higher frequencies and were far more accurate. One notable microwave radar developed at the time was by the MIT Radiation Laboratory named the SCR-584. This was a gunfire-control system that could place antiaircraft guns on target without the need for searchlights and optics, as was required for older models such as the SCR-268. It operated in the frequency



Exterior view of the SCR-584

range of 2.7 – 2.9 GHz (known as the S band). The system was first used in combat in 1944 in Italy and the use of this microwave radar helped Italy catch the Germans unprepared.

PROGRESS POST WAR

The 1950s saw new and better radar systems emerging. Large, high-powered radars that operated at 220 MHz (VHF) and 450 MHz (UHF) also began appearing in numbers. Synthetic aperture radar systems began appearing in the early 1950s, but it took almost 30 more years for these systems to become reliable. Prevalent application of the Doppler principle in radar began in the 1950s. Today the principle is vital in the operation of many radar systems.

In the 1970s, digital technologies were advancing rapidly which allowed for modern radar systems that required advanced signal and data processing. Advances were made in the airborne pulse Doppler radar which enhanced its ability to detect aircraft even in conditions with heavy ground clutter. Advances in remote sensing made it possible to measure winds over the sea, the geoid (or mean sea level), ocean roughness, and more with radar.

Advances in computer technology in the 1990s allowed for increased information about the targets to be passed on. Several new-age radar systems were now used in fields such as civil aviation, medicine, meteorology, and marine navigation. Further advances in digital technology in the 21st century led to the development of all-digital phased-array radars. 